Experiment (3)
Diode Characteristics

Objectives

- To experimentally determine the IV characteristics of a general purpose diode.
- To analyze a simple diode circuit using graphical methods.
- To study the application of diodes in logic gate design.

Pre-lab Work

Analyze the two circuits using the constant voltage diode model (V_{DO}=0.7 V).

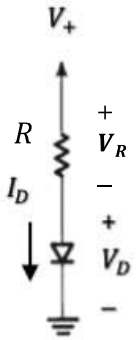


Figure 1: Diode IV curve.

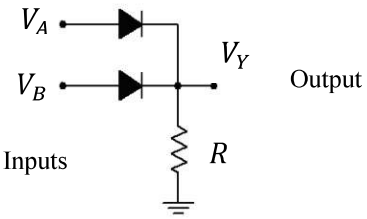


Figure 2: A diode logic gate circuit.

Solution

Equipment

Table 1. Equipment for experiment 3.

	Equipment	Model	Serial No.	Stock No.
1	BreadBoard Station			
2	1 kΩ Resistor (2W, 5%)			
3	Silicon Diodes (2)	1N4001	https://www.onsemi.com/pub/Collateral/1N4001-D.PDF	
4	Multi-meter (1)			
5	Probes and cables			

Experimental Procedure

Part (1): Diode Forward IV Characteristics.

1. Start first by checking if the diode is not damaged. Set the multimeter on resistance measurement. Touch the black lead (negative) of the multimeter to the cathode of the diode (refer to figure 3) and the red lead (positive) to the anode. In this configuration, the diode is forward-biased, and you should get a resistance reading between 1 kΩ and 10 MΩ. Next, switch the leads to the opposite terminals. The diode is now reverse-biased, and the reading should be infinity. If the readings are the same in both directions, the diode is damaged.

2. Connect the circuit in figure 1, with R=1 kΩ. Pay a special attention to how you connect the two terminals of the diode by referring to figure 3.
3. Starting with 0 volts, increase the DC voltage source V_+ up to 2 volts in the steps given in table 2. In each setting, measure the voltage across the diode and across the resistor using the multi-meter. Enter your measurements in table 2.

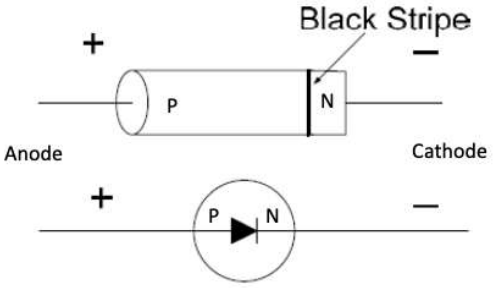


Figure 3: Diode marking.

Part (2): Graphical Analysis of Diode Circuits.

1. Consider again the circuit in figure 1, with a R=1 kΩ. and V_+=2 V.
2. Measure the voltages V_D and V_R using the multi-meter. Enter your measurements in table 3.
3. With the diode removed (open-circuit), measure the voltages V_D and V_R. Enter your measurements in table 3.
4. With the diode shorted, measure the voltages V_D and V_R using the multi-meter. Enter measurements in table 3.

Part (3): Diode Application as a Logical Gate.

1. Connect the circuit shown in figure 2.
2. Connect the inputs V_A and V_B to 0 volts or +5V for a total of four combinations, as shown in table 5. In each combination, record the value of the output voltage V_Y.

Results and Analysis

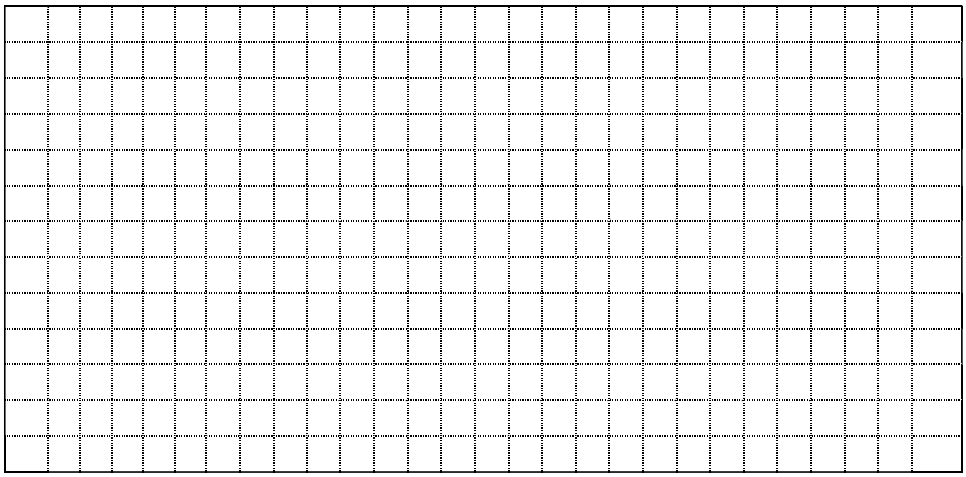
Part (1): Diode Forward IV Characteristics.

1. Calculate the diode current (I_D) in table 2.
2. Now, use the graphing space below to plot the diode characteristics I_D (vertical axis) versus V_R (horizontal axis). Appropriately label your plot.
3. Does the curve resemble the usual diode IV curve? Explain any discrepancies.

Answer:

Table 2. Measurements for Part (1).

DC Input V_+ (V)	V_D (V)	V_R (V)	$I_D = V_R/R$ (mA)
0.0			
0.3	0.2509	0.0001	0.0001
0.5	0.4066	0.0067	0.0067
0.6	0.4842	0.0653	0.0653
0.7	0.5230	0.1630	0.1630
0.8	0.5300	0.2473	0.2473
0.9	0.5541	0.3319	0.3319
1.0	0.5640	0.4140	0.4140
1.2	0.5700	0.6180	0.6180
1.6	0.6026	0.9548	0.9548
2.0	0.6182	1.4062	1.4062



Graph 1: Diode IV characteristics.

Part (2): Graphical Analysis of Diode Circuits.

1. Calculate the diode current (I_D) in table 3 for all three cases.
2. Now, on the same graphing space above (graph 1), draw the "load line" of the circuit. Use the second and third entries in table 3.
3. The intersection point between the load line and the diode characteristics curve is the "operating point" of the circuit. Find this point from the graph and record it in table 4.
4. Also, record in table 4 the experimental (table 3, first entry) and the theoretical values of the current in the circuit.
5. Calculate the percent differences. Which value do you think is more accurate? Explain.

Answer:

Table 3. Measurements for Part (2).

Diode	V_D (V)	V_R (V)	$I_D = V_R/R$ (mA)
Included	0.6180	1.3820	1.3820
Removed	0	0	0
Shorted	0.0004	1.9820	1.9820

Table 4. Calculations for Part (2).

I_D (mA) Experimental value	I_D (mA) Graphical value	% Difference	I_D (mA) Theoretical value	% Difference

Part (3): Diode Application as a Logical Gate.

1. From the results in table 5, what logic function does the circuit perform?
2. Diodes are not commonly used in the design of logic gate. Investigate and explain why not?

Answer: The curcuit does the OR logic function. While diode logic has the advantage of simplicity in some circuits, the absence of a boost stage at each gate limits its application. In addition, not all logical functions can be implemented using diode logic alone, making it less preferable because only non-inverting logical AND and logical OR functions can be realized with diode gates. Also, having several diode logic gates cascaded causes the voltage levels at each stage to vary considerably, which is not desirable.

Table 5. Measurements for Part (3).

V_A (V)	V_B (V)	V_Y (V)
0	0	0
0	5	4.2790
5	0	4.3267
5	5	4.3

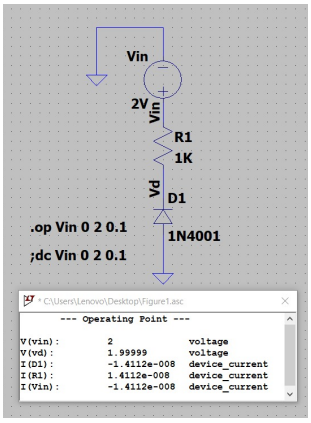
Additionally, the diode logic is normally limited to a single stage, but in special designs two-stage systems are sometimes obtained. We also used two 0.7 volt forward bias diodes in our experiment, and when we applied a potential of 5 volts to the other, where a potential did not pass through, we obtained a value close to 4.3 volts.

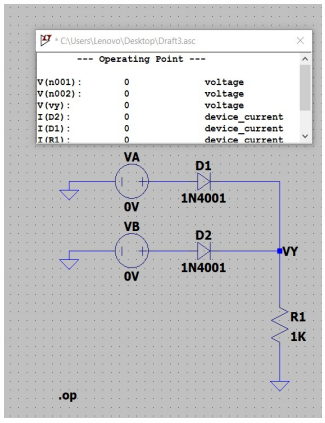
LTspice Simulations

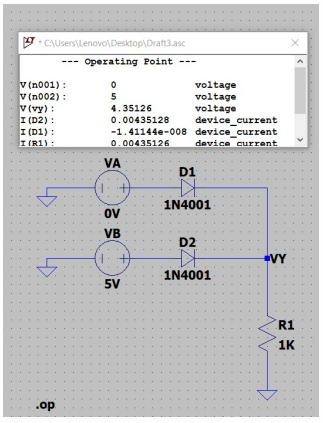
Include with your report LTspice simulation results of all parts of this experiment.

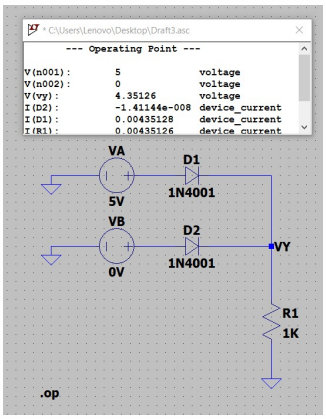
Conclusions and Remarks

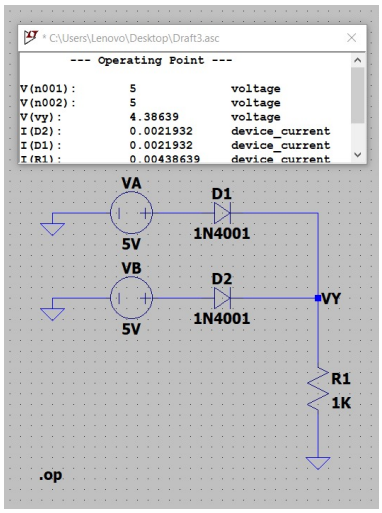
Write below all your conclusions/remarks on this experiment.

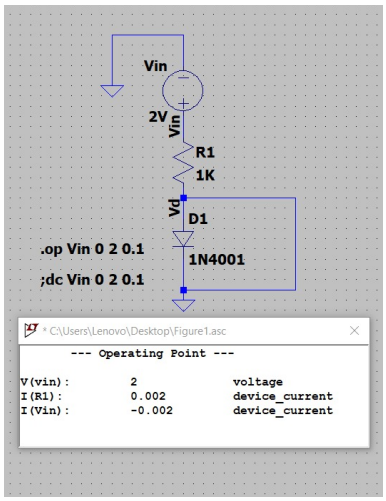


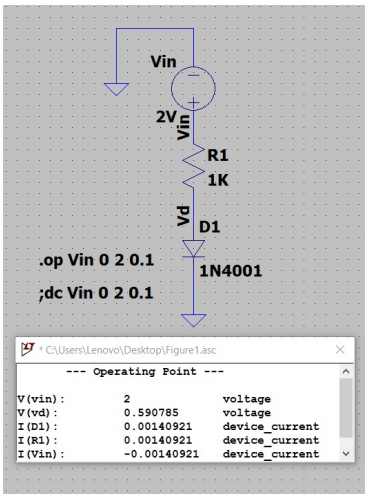












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FAIRCHILD

November 2014

1N4001 - 1N4007
General-Purpose Rectifiers

Features

- Low Forward Voltage Drop
- High Surge Current Capability



DO-41
COLOR BAND DENOTES CATHODE

Ordering Information

Part Number	Top Mark	Package	Packing Method
1N4001	1N4001	DO-204AL (DO-41)	Tape and Reel
1N4002	1N4002	DO-204AL (DO-41)	Tape and Reel
1N4003	1N4003	DO-204AL (DO-41)	Tape and Reel
1N4004	1N4004	DO-204AL (DO-41)	Tape and Reel
1N4005	1N4005	DO-204AL (DO-41)	Tape and Reel
1N4006	1N4006	DO-204AL (DO-41)	Tape and Reel
1N4007	1N4007	DO-204AL (DO-41)	Tape and Reel

Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only. Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter	Value							Unit
		1N 4001	1N 4002	1N 4003	1N 4004	1N 4005	1N 4006	1N 4007	
V_{RRM}	Peak Repetitive Reverse Voltage	50	100	200	400	600	800	1000	V
$I_{F(AV)}$	Average Rectified Forward Current .375 " Lead Length at T_A = 75°C	1.0							A
I_{FSM}	Non-Repetitive Peak Forward Surge Current 8.3 ms Single Half-Sine-Wave	30							A
I^2t	Rating for Fusing (t < 8.3 ms)	3.7							A^2sec
T_{STG}	Storage Temperature Range	-55 to +175							°C
T_J	Operating Junction Temperature	-55 to +175							°C

Thermal Characteristics

Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter	Value	Unit
P_D	Power Dissipation	3.0	W
$R_{\theta JA}$	Thermal Resistance, Junction-to-Ambient	50	°C/W

Electrical Characteristics

Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter	Conditions	Value	Unit
V_F	Forward Voltage	I_F = 1.0 A	1.1	V
I_{rr}	Maximum Full Load Reverse Current, Full Cycle	T_A = 75°C	30	µA
I_R	Reverse Current at Rated V_R	T_A = 25°C	5.0	µA
		T_A = 100°C	50	
C_T	Total Capacitance	V_R = 4.0 V, f = 1.0 MHz	15	pF